

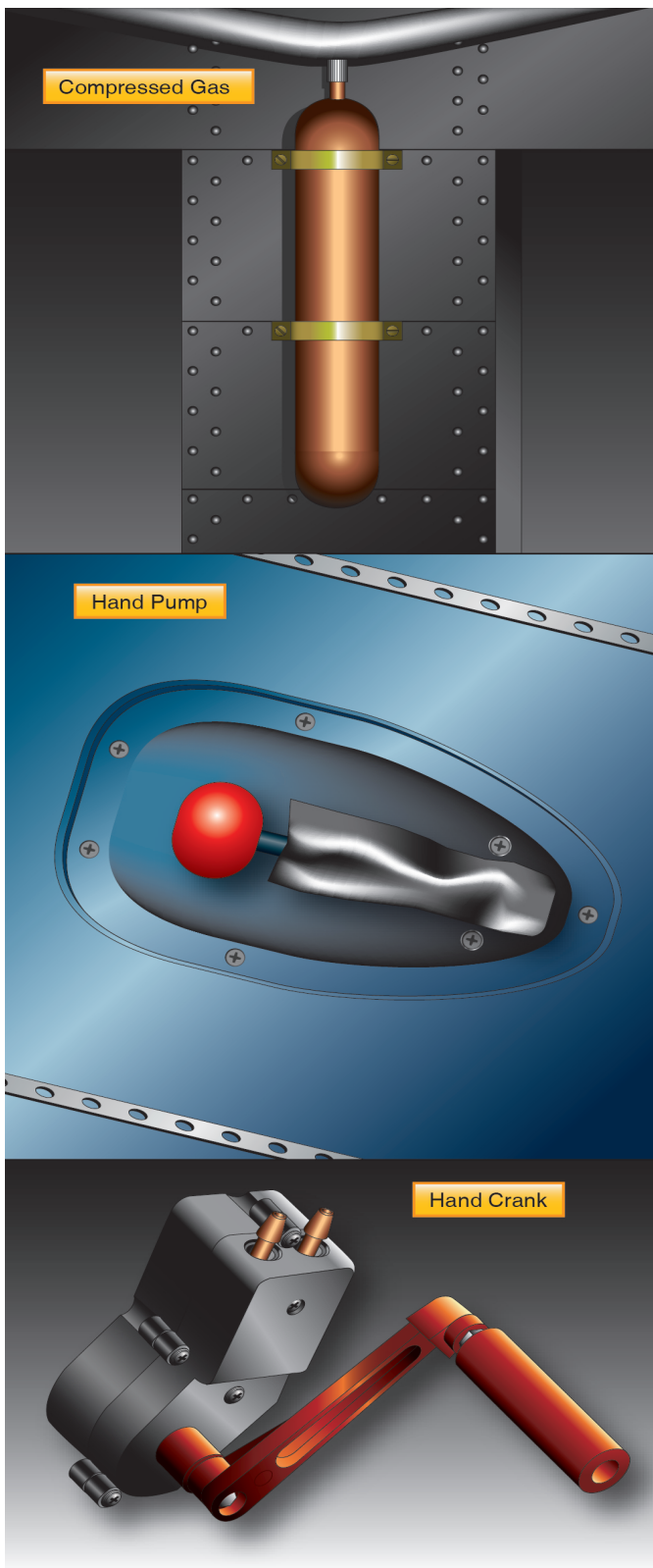


Figure 12-13. Typical emergency gear extension systems.

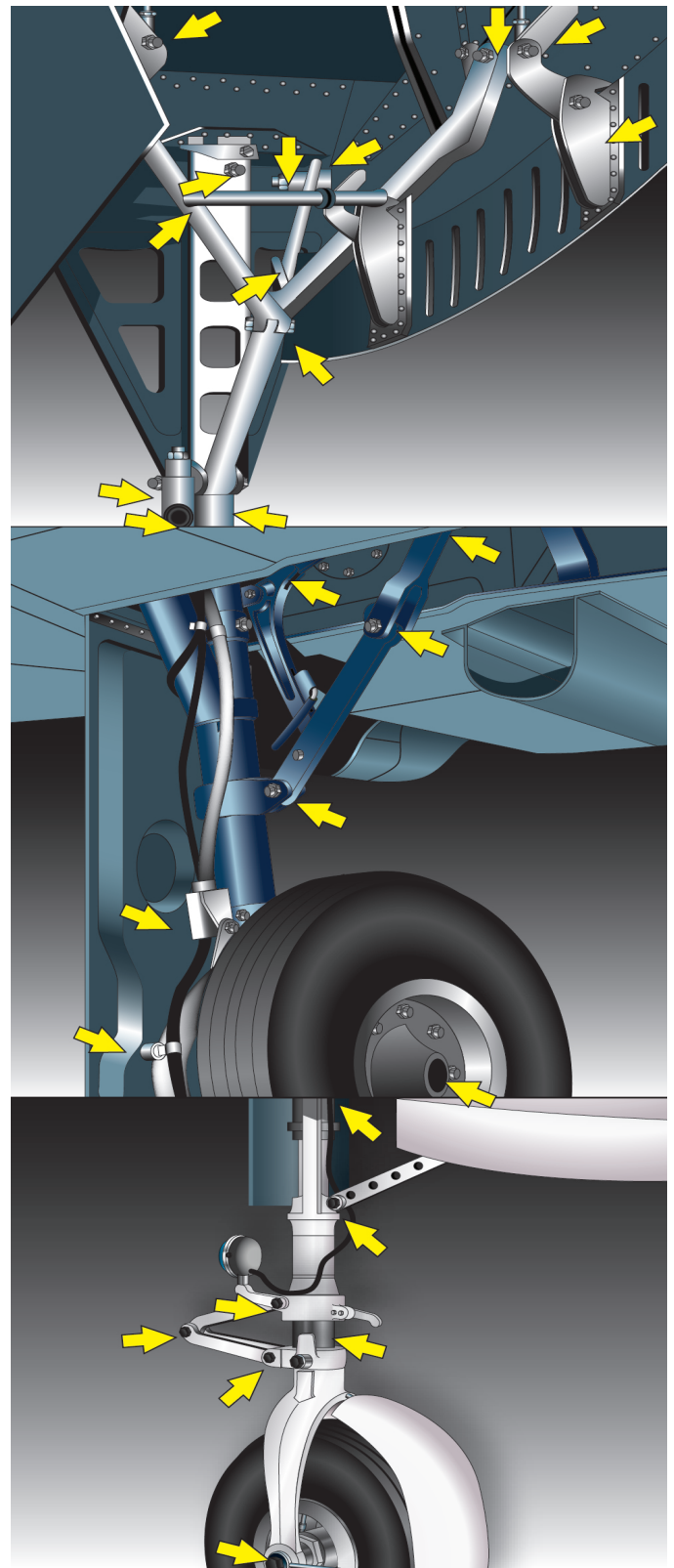


Figure 12-14. Retractable landing gear inspection checkpoints.

External inspection of the landing gear consists of checking individual system components. [Figure 12-14] The landing gear, wheel well, and adjacent areas should be clean and free of mud and debris. Dirty switches and valves may cause false safe light indications or interrupt the extension cycle before the landing gear is completely down and locked. The wheel wells should be clear of any obstructions, as foreign objects may damage the gear or interfere with its operation. Bent gear doors may be an indication of possible problems with normal gear operation.

Ensure shock struts are properly inflated and that the pistons are clean. Check main gear and nose gear uplock and downlock mechanisms for general condition. Power sources and retracting mechanisms are checked for general condition, obvious defects, and security of attachment. Check hydraulic lines for signs of chafing and leakage at attach points. Warning system micro switches (squat switches) are checked for cleanliness and security of attachment. Actuating cylinders, sprockets, universal joints, drive gears, linkages, and any other accessible components are checked for condition and obvious defects. The airplane structure to which the landing gear is attached is checked for distortion, cracks, and general condition. All bolts and rivets should be intact and secure.

Takeoff and Climb

Normally, the landing gear is retracted after lift-off when the airplane has reached an altitude where, in the event of an engine failure or other emergency requiring an aborted takeoff, the airplane could no longer be landed on the runway. This procedure, however, may not apply to all situations. Preplan landing gear retraction taking into account the following:

- Length of the runway
- Climb gradient
- Obstacle clearance requirements
- The characteristics of the terrain beyond the departure end of the runway
- The climb characteristics of the particular airplane

For example, in some situations it may be preferable, in the event of an engine failure, to make an off airport forced landing with the gear extended in order to take advantage of the energy absorbing qualities of the terrain (see Chapter 18, “Emergency Procedures”). In which case, a delay in retracting the landing gear after takeoff from a short runway may be warranted. In other situations, obstacles in the climb path may warrant a timely gear retraction after takeoff. Also, in some airplanes the initial climb pitch attitude is such that any view of the runway remaining is blocked, making an assessment of the feasibility of touching down on the remaining runway difficult.

Avoid premature landing gear retraction and do not retract the landing gear until a positive rate of climb is indicated on the flight instruments. If the airplane has not attained a positive rate of climb, there is always the chance it may settle back onto the runway with the gear retracted. This is especially so in cases of premature lift-off. Remember that leaning forward to reach the landing gear selector may result in inadvertent forward pressure on the yoke, which causes the airplane to descend.

As the landing gear retracts, airspeed increases and the airplane’s pitch attitude may change. The gear may take several seconds to retract. Gear retraction and locking (and gear extension and locking) is accompanied by sound and feel that are unique to the specific make and model airplane. Become familiar with the sound and feel of normal gear retraction so that any abnormal gear operation can be readily recognized. Abnormal landing gear retraction is most often a clear sign that the gear extension cycle will also be abnormal.

Approach and Landing

The operating loads placed on the landing gear at higher airspeeds may cause structural damage due to the forces of the airstream. Limiting speeds, therefore, are established for gear operation to protect the gear components from becoming overstressed during flight. These speeds may not be found on the airspeed indicator. They are published in the AFM/POH for the particular airplane and are usually listed on placards in the flight deck. [Figure 12-15] The maximum landing extended speed (V_{LE}) is the maximum speed at which the airplane can be flown with the landing gear extended. The maximum landing gear operating speed (V_{LO}) is the maximum speed at which the landing gear may be operated through its cycle.



Figure 12-15. *Placarded gear speeds in the flight deck.*

The landing gear is extended by placing the gear selector switch in the GEAR DOWN position. As the landing gear extends, the airspeed decreases and the pitch attitude may change. During the several seconds it takes for the gear to extend, be attentive to any abnormal sounds or feel. Confirm that the landing gear has extended and locked by the normal sound and feel of the system operation, as well as by the gear position indicators in the flight deck. Unless the landing gear has been previously extended to aid in a descent to traffic pattern altitude, the landing gear should be extended by the time the airplane reaches a point on the downwind leg that is opposite the point of intended landing. Establish a standard procedure consisting of a specific position on the downwind leg at which to lower the landing gear. Strict adherence to this procedure aids in avoiding unintentional gear up landings.

Operation of an airplane equipped with a retractable landing gear requires the deliberate, careful, and continued use of an appropriate checklist. When on the downwind leg, make it a habit to complete the before-landing checklist for that airplane. This accomplishes two purposes—it ensures that action has been taken to lower the gear and establishes awareness so that the gear down indicators can be rechecked prior to landing.

Unless good operating practices dictate otherwise, the landing roll should be completed and the airplane should be clear of the runway before any levers or switches are operated. This technique greatly reduces the chance of inadvertently retracting the landing gear while on the ground. Wait until after rollout and clearing the runway to focus attention on the after-landing checklist. This practice allows for positive identification of the proper controls.

When transitioning to retractable gear airplanes, it is important to consider some frequent pilot errors. These include pilots that have:

- Neglected to extend landing gear
- Inadvertently retracted landing gear
- Activated gear but failed to check gear position
- Misused emergency gear system
- Retracted gear prematurely on takeoff
- Extended gear too late

These mistakes are not only committed by pilots who have just transitioned to complex aircraft, but also by pilots who have developed a sense of complacency over time. In order to minimize the chances of a landing gear-related mishap:

- Use an appropriate checklist. (A condensed checklist mounted in view is a reminder for its use and easy reference can be especially helpful.)
- Be familiar with, and periodically review, the landing gear emergency extension procedures for the particular airplane.
- Be familiar with the landing gear warning horn and warning light systems for the particular airplane. Use the horn system to cross-check the warning light system when an unsafe condition is noted.
- Review the procedure for replacing light bulbs in the landing gear warning light displays for the particular airplane, if applicable, so that you can properly replace a bulb to determine if the bulb(s) in the display is good. If bulbs are replaceable, check to see if spare bulbs are available in the airplane spare bulb supply as part of the preflight inspection.
- Be familiar with and aware of the sounds and feel of a properly operating landing gear system.

Transition Training

Transition to a complex airplane or a high-performance airplane should be accomplished through a structured course of training administered by a competent and qualified flight instructor. The training should be accomplished in accordance with a ground and flight training syllabus. *[Figure 12-16]*

This sample syllabus for transition training is an example. The arrangement of the subject matter may be changed and the emphasis shifted to fit the qualifications of the transitioning pilot, the airplane involved, and the circumstances of the training situation. The goal is to ensure proficiency standards are achieved. These standards are contained in the Airman Certification Standards for the certificate that the transitioning pilot holds or is working toward.

Ground Instruction	Flight Instruction
One hour	One hour
1. Operations sections of flight manual	1. Flight training maneuvers
2. Line inspection	2. Takeoffs, landings and go-arounds
3. Flight deck familiarization	
One hour	One hour
1. Aircraft loading, limitations and servicing	1. Emergency operations
2. Instruments, radio and special equipment	2. Control by reference to instruments
3. Aircraft systems	3. Use of radio and autopilot
One hour	One hour
1. Performance section of flight manual	1. Short and soft-field takeoffs and landings
2. Cruise control	2. Maximum performance operations
3. Review	

Figure 12-16. *Sample transition training syllabus.*

The training times indicated in the syllabus are for illustration purposes. Actual times should be based on the capabilities of the pilot. The time periods may be minimal for pilots with higher qualifications or increased for pilots who do not meet certification requirements or have had little recent flight experience.

Chapter Summary

Flying a complex or high-performance airplane requires a pilot to further divide his or her attention during the most critical phases of flight: takeoff and landing. The knowledge, judgment, and piloting skills required to fly these airplanes needs to be developed. It is essential that adequate training is received to ensure a complete understanding of the systems, their operation (both normal and emergency), and operating limitations.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 13: Transition to Multiengine Airplanes

Introduction

This chapter is devoted to the factors associated with the operation of small multiengine airplanes. For the purpose of this handbook, a “small” multiengine airplane is a reciprocating or turbopropeller-powered airplane with a maximum certificated takeoff weight of 12,500 pounds or less. This discussion assumes a conventional design with two engines—one mounted on each wing. Reciprocating engines are assumed unless otherwise noted. The term “light-twin,” although not formally defined in the regulations, is used herein as a small multiengine airplane with a maximum certificated takeoff weight of 6,000 pounds or less.

There are several unique characteristics of multiengine airplanes that make them worthy of a separate class rating. The one engine inoperative (OEI) flight information presented in this chapter emphasizes the significant difference between flying a multiengine and a single-engine airplane. However, all pilots need appropriate knowledge, risk management strategies, and skills to fly safely in any airplane they fly, and mastery of OEI flight is only one aspect of safe multiengine flying. The modern, well-equipped multiengine airplane can be remarkably capable under many circumstances, but, the performance and system redundancy of a multiengine airplane only increase safety if the pilot is trained and proficient.

The airplane manufacturer is the final authority on the operation of a particular make and model airplane. Flight instructors and learners should use the Federal Aviation Administration’s Approved Flight Manual (AFM) and/or the Pilot’s Operating Handbook (POH). The airplane manufacturer’s guidance and procedures take precedence over any general recommendations made in this handbook.

General

Multiengine and single-engine airplanes operate differently during an engine failure. In a multiengine airplane, loss of thrust from one engine affects both *performance and control*. The most obvious problem is the loss of 50 percent of power, which reduces climb performance 80 to 90 percent. In some cases after an engine failure, the ability to climb or maintain altitude in a light-twin may not exist. After an engine failure, asymmetrical thrust also creates control issues for the pilot. Attention to both these factors is crucial to safe OEI flight.

Terms and Definitions

Pilots of single-engine airplanes are already familiar with many performance “V” speeds and their definitions. Twin-engine airplanes have several additional V-speeds unique to OEI operation. These speeds are differentiated by the notation “SE” for single engine. A review of some key V-speeds and several new V-speeds unique to twin-engine airplanes are listed below.

- V_R —rotation speed—speed at which back pressure is applied to rotate the airplane to a takeoff attitude.
- V_{LOF} —lift-off speed—speed at which the airplane leaves the surface. (Note: Some manufacturers reference takeoff performance data to V_R , others to V_{LOF} .)
- V_X —best angle of climb speed—speed at which the airplane gains the greatest altitude for a given distance of forward travel.
- V_{XSE} —best angle-of-climb speed with OEI.
- V_Y —best rate of climb speed—speed at which the airplane gains the most altitude for a given unit of time.
- V_{YSE} —best rate of climb speed with OEI. Marked with a blue radial line on most airspeed indicators. Above the single-engine absolute ceiling, V_{YSE} yields the minimum rate of sink.
- V_{SSE} —safe, intentional OEI speed—originally known as safe single-engine speed. It is the minimum speed to intentionally render the critical engine inoperative.
- V_{REF} —reference landing speed—an airspeed used for final approach, which is normally 1.3 times V_{SO} , the stall speed in the landing configuration. The pilot may adjust the approach speed for winds and gusty conditions by using V_{REF} plus an additional number of units (e.g., $V_{REF}+5$).